

Association between triglyceride-glucose index and chronic kidney disease: A cohort study and meta-analysis

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KEYWORDS

Triglyceride-glucose index;
TyG index;
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meta-Analysis

Abstract *Background and aims:* Nowadays, the relationship between triglyceride-glucose (TyG) index and chronic kidney disease (CKD) is still controversial. We aimed to prospectively investigate the relationship between TyG index and CKD in a cohort and meta-analysis.

Methods and results: A total of 10498 participants from the China Health and Retirement Longitudinal Study (CHARLS) were included. Participants were divided into four groups based on the quartiles of the TyG index. CKD was based on self-reported physicians' diagnosis or personal eGFR level. A cox regression model was established to analyze the correlation between TyG index and CKD. A meta-analysis was conducted to incorporate the results of the current study and previous studies on the association of TyG index with CKD. In multivariable-adjusted analyses, the adjusted hazard ratio (95% confidence interval) for the highest versus lowest quartile of TyG index was 1.30 (1.08–1.57). Each 1-SD higher TyG index was associated with an increased risk of 11% (HR 1.11, 95% CI 1.03–1.19). The meta-analysis further confirmed the significant associations between TyG and CKD and pooled relative risk for highest vs lowest TyG index quartile was 1.47 (1.32–1.63).

Conclusions: Higher TyG index was associated with increased risk of CKD, independently of established risk factors. The TyG index may be a predictor of incident CKD.

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Abbreviations: TyG, triglyceride-glucose; CKD, chronic kidney disease; GBD, Global Burden of Disease; IR, insulin resistance; CHARLS, China Health and Retirement Longitudinal Study; FBG, fasting blood glucose; Scr, serum creatinine; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure; HR, hazard ratio; RR, relative risk; OR, odd ratio; CI, confidence intervals; NOS, Newcastle–Ottawa Scale; MAU, microalbuminuria; ABI, ankle-brachial index; HDL, high-density lipoprotein; ROS, reactive oxidative stress.

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1. Introduction

Chronic kidney disease (CKD) has been increasingly a prevalent health concern and its progression often leads to severe adverse outcomes, including decreased renal function, renal failure, increased the incidence of end-stage renal disease, cardiovascular disease and other related complications [1–4], which will lead to an increase in the burden on the economy and medical resources [5]. With an average of over 21 million cases of CKD per year, nearly 1.2 million deaths from CKD, and 35 million years of healthy life lost due to CKD [6], CKD was ranked as the 12th leading cause of death among 133 diseases by Global Burden of Disease (GBD) [7,8]. Survey showed that the prevalence of

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CKD in Chinese adults is 10.8% [9,10]. Therefore, early diagnosis and intervention are essential to reduce the incidence of CKD and the burden caused by poor prognosis.

Insulin resistance (IR) is defined clinically as the inability of a known quantity of exogenous or endogenous insulin to increase glucose uptake and utilization in an individual as much as it does in a normal population [11], which is a risk factor for the progression of CKD. The hyperinsulinemic-euglycemic clamp technique is the gold standard for evaluating IR, not commonly used in large-scale epidemiological studies due to its high cost and invasiveness [12]. Recently, some surrogate indicators of IR have been used widely, such as the Triglyceride-Glucose (TyG) index (In 2008, Simental-Mendía et al. first proposed the TyG index as a surrogate index for identifying IR). Previous studies have shown that the TyG index has high sensitivity and specificity for the diagnosis of insulin resistance [13], and has the advantages of low cost, easy measurement, and good social benefits.

Previous studies have shown that insulin resistance, prediabetes and diabetes are risk factors for CKD, and the TyG index can reflect insulin resistance to a certain extent. Therefore, some studies have evaluated the relationship between the TyG index as a surrogate index of insulin resistance and CKD, but the results of studies have obvious inconsistencies. Some studies have shown a significant relationship between TyG index and CKD [14–24], but Pan et al. found no significant correlation between TyG index and CKD [25]. Different results may be due to group differences represented by participants, inclusion criteria and type of study design. In order to investigate the relationship between TyG index and CKD, we conducted a large-scale cohort study to test the association between the TyG index and CKD based on the China Health and Retirement Longitudinal Study (CHARLS), and combined the previous studies to carry out a meta-analysis to further verify the conclusions.

2. Methods

2.1. Study population

This study was conducted based on CHARLS, which is a longitudinal study of families and individuals aged 45 and above in China. CHARLS National Baseline Survey was conducted in 2011, which used a multistage clustering sample method to select participants in China. The baseline sample included 17,708 respondents in 10,257 households in 450 villages/urban communities in 150 counties/districts in 28 provinces. Then, three follow-up visits were followed from July 2013 to January 2014 (Wave 2), July 2015 to January 2016 (Wave 3) and July to November 2018 (Wave 4) among surviving participants.

A total of 17,708 participants were included in the baseline survey in 2011, and those who met the following conditions will be included in this study: aged 45 years and older; complete information on triglyceride (TG), fasting blood glucose (FBG) and serum creatinine (Scr); without history of CKD at the baseline survey (Wave 1). A

total of 10,498 participants were eventually included in this analysis (Figure S1).

2.2. Sample collection

Well-trained Chinese Center for Disease Control and Prevention staff were responsible for collecting venous blood samples from participants who fasted at night, and promptly transported the samples to the local laboratory for storage at 4 °C. Then the venous blood was separated into plasma and buffy coat, which were immediately stored frozen at –20 °C and transported to the Chinese CDC in Beijing within 2 weeks where they were placed in a deep freezer and stored at –80 °C until assay at CMU laboratory. The enzymatic colorimetric test method was used to measure TG and FBG, rate-blanked and compensated Jaffe creatinine method was used to measure Scr.

2.3. Definition of TyG index and CKD

The TyG index was calculated as $\ln [\text{fasting triglycerides (mg/dl)} \times \text{fasting glucose (mg/dl)} / 2]$ [26]. CKD was based on self-reported physicians' diagnosis ('Have you ever been told by a doctor that you have kidney diseases?') or personal estimate glomerular filtration rate (eGFR) level from Wave 2 to Wave 4 (The eGFR was calculated from the Epidemiology Collaboration (CKD-EPI) equation [27]). If an affirmative answer was given by an individual or his/her proxy respondent or $\text{eGFR} < 60 \text{ mL/min/1.73 m}^2$, then the individual would be categorized as experiencing the first-time CKD in his/her life (except for tumor or cancer) [28].

2.4. Covariates assessments

The covariates were collected at baseline including age, sex, place of residence (urban vs. rural), educational level (below primary school; primary school; middle school; high school or above), chronic diseases history (hypertension, dyslipidemia, diabetes mellitus, cancer, chronic lung disease, stroke, liver disease, arthritis, digestive disease and asthma), smoking status (ever smoking vs. never smoking), drinking status (ever drinking vs. never drinking), body mass index (BMI), systolic blood pressure (SBP) and diastolic blood pressure (DBP). BMI was defined as weight in kilograms divided by height in meters squared. Blood pressure was defined as the average of three separate blood pressure measurements for participants using an electronic sphygmomanometer (OmronHEM-7200 monitor) after a 5 min rest in the seated position. Hypertension was defined as systolic blood pressure $\geq 140 \text{ mm Hg}$, diastolic blood pressure $\geq 90 \text{ mm Hg}$, current use of antihypertensive medications, or self-reported history of hypertension. Dyslipidemia was defined as triglycerides $\geq 150 \text{ mg/dL}$, total cholesterol $\geq 240 \text{ mg/dL}$, high-density lipoprotein cholesterol $< 40 \text{ mg/dL}$, low-density lipoproteins cholesterol $\geq 160 \text{ mg/dL}$, current use of the lipid-lowering medications, or self-reported history of dyslipidemia. Diabetes mellitus was defined as fasting glucose $\geq 126 \text{ mg/dL}$, glycosylated hemoglobin (HbA1c) $\geq 6.5\%$,

treatment for diabetes mellitus, or self-reported history of diabetes.

2.5. Statistical analysis

All participants were categorized into four groups according to the quartiles of TyG index. Demographic and clinical characteristics were compared between four groups by ANOVA or Kruskal–Wallis test for continuous variables which were expressed as n (%) and χ^2 test for categorical variables. In this study, a Cox regression model was generated to determine the association between TyG index and CKD. The hazard ratios (HRs) and 95% confidence intervals (CIs) were estimated for higher quartiles compared with the lowest quartile and for per SD increment of TyG index (0.76) [29]. Potential covariates, such as age, sex, place of residence, educational level, chronic diseases history (hypertension, dyslipidemia, diabetes mellitus, cancer, chronic lung disease, stroke, liver disease, arthritis, digestive disease, asthma), smoking status, drinking status, SBP, DBP and if use of antihypertensive, lipid-lowering and anti-diabetic medications. Additionally, a spline regression model was used to describe the shape of association between TyG index and CKD, fitting a restricted cubic spline function with four knots (at the 5th, 35th, 65th, and 95th percentiles). Subgroup analyses were further performed to evaluate the association between TyG index and the risk of CVD according to sex, age, place of residence, BMI, smoking, drinking, hypertension, and educational level. A two-sided $P < 0.05$ was considered statistically significant. All statistical analyses were performed using SAS statistical software version 9.4 (SAS Institute, Cary, NC, USA).

2.6. Meta-analysis

We combined this study and previously published studies on the association of TyG index with CKD conducting a meta-analysis according to the Preferred Systematic Review and Reporting Project for Meta-Analysis (PRISMA) guidelines [30]. Literature search and data extraction were performed independently by 2 investigators (Xiaowei Zheng and Xiao Ren). A systematic literature search was conducted on PubMed and Web of Science from inception to March 2023, using the search terms ‘triglyceride-glucose index’ (or ‘TyG index’, ‘triglyceride-glucosew’) in combined with ‘chronic kidney diseases’ (or ‘chronic renal insufficiencies’, ‘chronic kidney insufficiency’, ‘chronic renal disease’, ‘CKD’). We applied no publication or language restrictions. The reference lists of previously published reviews and meta-analyses, and original research articles were additionally scrutinised to identify publications not covered by the original database searches. Studies were selected if they met the following criteria: (a) included an investigation of the association between TyG index and CKD; (b) study design was a cross-sectional study or cohort study; (c) reported covariate adjusted hazard ratios (HRs), relative risks (RRs), odds ratios (ORs) with 95% confidence intervals (CIs) or valid data can be

provided in the literature to calculate the HRs, RRs or ORs. Quality assessment was conducted according to the Newcastle–Ottawa Scale (NOS) criteria for non-randomized studies [31]. The following data elements were extracted from each included study: first author, year and place of study, type of study design, number of participants, participants’ age, effect sizes (HRs, relative risks (RRs) or odd ratios (ORs)), and adjusted covariates variable. Cochrane’s Q test and I^2 statistic were used to assess the heterogeneity among the results of the studies. Funnel plot, Begg’s test (rank correlation test) and Egger’s test (linear regression method) were used to evaluate the possibility of publication bias in the included literature. All data were analyzed using STATA version 16.0 (Stata Corp LP, College Station, TX, USA). Identically, the two-sided $P < 0.05$ was considered statistically significant. The detailed methods of this meta-analysis are described in Data S1.

3. Results

3.1. Baseline characteristics of study participants

In the present study, 10,498 participants (4935 men and 5563 women) were included in the analysis, and the average age was 59.00 ± 9.47 years. The quartiles for TyG were as follows: Q1 < 8.23 ; Q2 8.23–9.61; Q3 8.61–9.07; Q4 ≥ 9.07 . Baseline characteristics, such as place of residence, history of dyslipidemia and diabetes mellitus, smoking, drinking, body mass index, systolic and diastolic blood pressure were significantly different among the four subgroups (Table 1).

3.2. Association between TyG index and CKD

Kaplan–Meier curves showed that the cumulative incidence rate of the highest TyG index group (Q4) was higher than the lowest TyG index group (Q1) at any follow-up time (Figure S2) with increasing follow-up time. The prevalence of CKD from Q1 to Q4 was 8.57%, 8.83%, 9.14%, and 10.28%. In the unadjusted model, comparing with the lowest TyG index quartile, the HR (95% CI) for the highest TyG index quartile was 1.22 (1.02–1.47). In the multivariable-adjusted model, the HR (95% CI) for the highest TyG index quartile was 1.30 (1.08–1.57). Each 1-SD higher TyG index was associated with an increased risk of 11% (HR 1.11, 95% CI 1.03–1.19) (Table 2). Multivariable-adjusted restricted cubic spline analyses presented linear association of TyG index with risk of CKD (p for linearity = 0.008) (Fig. 1).

In the subgroup analysis, significant associations of higher TyG index with risk of CKD were observed in sex, age, place of residence, smoking, hypertension and education level subgroups. However, no significant interaction between TyG index and CKD were observed (Table 3).

3.3. Meta-analysis

Twelve published articles and the current study were included in this meta-analysis. These studies were

Table 1 Baseline characteristics of the study participants according to TyG index quartiles at baseline.

Characteristics	TyG index				P trend
	Q1 (<8.23)	Q2 (8.23–8.61)	Q3 (8.61–9.07)	Q4 (≥9.07)	
No. of subjects	2615	2640	2627	2616	
Age, years	59.15 ± 9.93	58.92 ± 9.47	59.31 ± 9.32	58.63 ± 9.13	0.160
Place of residence, n (%)					
Urban	817 (31.24)	913 (34.58)	994 (36.23)	1108 (41.04)	<0.001
Rural	1798 (68.76)	1727 (65.42)	1633 (62.16)	1508 (57.65)	
Education level, n (%)					
Below primary school	736 (28.15)	776 (29.39)	823 (31.33)	748 (28.59)	0.204
Primary school	1099 (42.03)	1039 (39.36)	1027 (39.09)	1005 (38.42)	
Middle school	521 (19.92)	539 (20.42)	504 (19.09)	547 (20.91)	
High school or above	259 (9.90)	286 (10.83)	273 (10.39)	316 (12.08)	
Chronic diseases history					
Hypertension, n (%)	556 (21.26)	575 (21.78)	580 (22.08)	625 (23.89)	0.024
Dyslipidemia, n (%)	119 (4.55)	194 (7.35)	262 (9.97)	448 (17.13)	<0.001
Diabetes mellitus, n (%)	56 (2.14)	98 (3.71)	143 (5.44)	358 (13.69)	<0.001
Smoking, n (%)	1157 (44.24)	1023 (38.75)	960 (36.54)	939 (35.89)	<0.001
Drinking, n (%)	1129 (43.17)	1037 (39.28)	932 (35.48)	961 (36.74)	<0.001
BMI, kg/m ²	22.39 (20.20–23.56)	23.40 (21.01–24.38)	23.40 (21.66–25.74)	23.82 (23.06–26.82)	<0.001
SBP, mmHg	127.31 ± 19.57	128.71 ± 20.05	131.45 ± 21.72	133.06 ± 19.75	<0.001
DBP, mmHg	74.18 ± 11.33	75.08 ± 11.42	76.18 ± 11.23	77.49 ± 11.12	<0.001

BMI: body mass index; SBP: systolic blood pressure; DBP: diastolic blood pressure; Continuous variables are expressed as mean ± standard deviation, or as median (interquartile range). Categorical variables are expressed as frequency (percent).

published between 2019 and 2022, which made up of 5 cross-sectional [14,18,20,24,25], 7 cohort studies [15–17,19,21–23]. 8 of the included studies were performed in China [14,15,17–19,22,24,25] and 4 studies were conducted in other countries (including Japan [16], America [20], Singapore [21] and Iran [23]). The sample size ranged from 1571 to 13,055 for a total number of 70,279 (Table S1).

The summary relative risk for the highest TyG index vs the lowest TyG index was 1.47 (1.32–1.63) (Fig. 2). A substantial level of heterogeneity was observed among the various studies ($I^2 = 50.40\%$, $P = 0.011$). Begg's test and Egger's test suggested the possibility of publication bias, respectively. Sensitivity analyses further confirmed a significant association between TyG index and CKD (Table S2).

In addition, subgroup analyses according to the type of study design, region and gender confirmed a significant correlation between TyG index and CKD. The cohort study and cross-sectional study group suggested an association between TyG and CKD (the pooled relative risk of 1.47, 95%

CI: 1.35–1.59 and 1.55, 95% CI: 1.18–2.04), respectively. Both the China and other countries groups suggested a correlation between TyG index and CKD (the pooled relative risk of 1.53, 95% CI: 1.29–1.81 and 1.42, 95% CI: 1.26–1.60). Both the male and female groups suggested a correlation between TyG index and CKD (the pooled relative risk of 1.58, 95% CI: 1.28–1.95 and 1.58, 95% CI: 1.26–1.98).

4. Discussion

In this large-scale prospective study based on CHARLS, we found TyG index was significantly correlated with CKD risk, and elevated TyG index was an independent risk factor for increased risk of CKD. Furthermore, in the meta-analysis we confirmed that individuals with higher TyG index had an increased risk of CKD.

Previous studies reported that IR is related to high risk factors of CKD. Considering the inconvenience of IR in clinical application, some studies invented the TyG index and reported the relationship between the TyG index and

Table 2 Association between baseline TyG index and CKD.

Variable	TyG index				P trend	Each SD increase
	Q1 (<8.23)	Q2 (8.23–8.61)	Q3 (8.61–9.07)	Q4 (≥9.07)		
Case, n (%)	224 (8.57)	233 (8.83)	240 (9.14)	269 (10.28)		
Unadjusted	1.00(Ref)	1.03 (0.85–1.57)	1.07 (0.89–1.30)	1.22 (1.02–1.47)	0.031	1.09 (1.01–1.17)
Age and sex- adjusted	1.00(Ref)	1.06 (0.87–1.28)	1.11 (0.92–1.35)	1.27 (1.06–1.54)	0.001	1.10 (1.03–1.18)
Multivariable-adjusted	1.00(Ref)	1.06 (0.87–1.29)	1.11 (0.92–1.35)	1.30 (1.08–1.57)	0.006	1.11 (1.03–1.19)

Each SD increase in TyG index (0.76).

Multivariable-adjusted adjusted for age, sex, place of residence, education level, smoking, drinking, systolic blood pressure, diastolic blood pressure, chronic diseases (hypertension, dyslipidemia, diabetes mellitus, cancer, chronic lung disease, stroke, liver disease, arthritis, digestive disease, asthma).

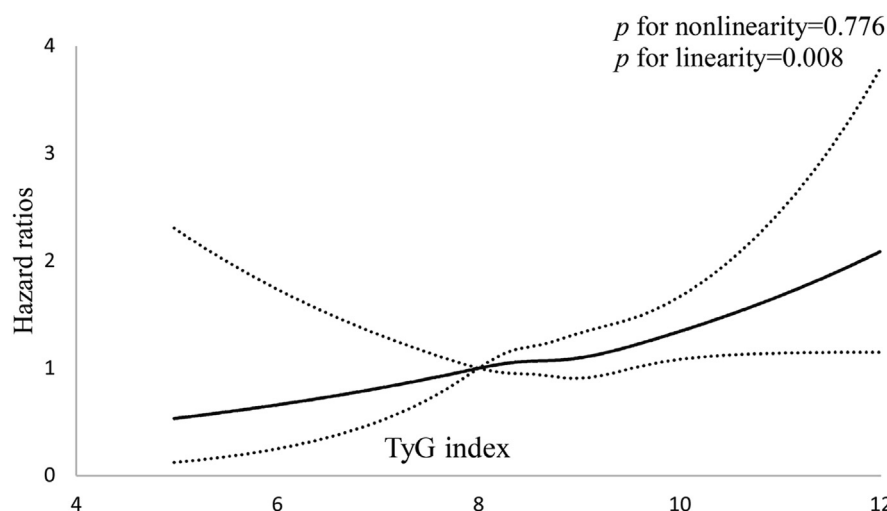


Figure 1 Association of TyG index with risk of CKD. Hazard ratios (HRs) and 95% confidence intervals were derived from restricted cubic spline regression, with knots placed at the 5th, 35th, 65th, and 95th percentiles of the distribution of TyG index. The reference point for TyG index is the midpoint (8.00) of the reference group from categorical analysis. HRs were adjusted for age, sex, place of residence, education level, smoking, drinking, systolic blood pressure, diastolic blood pressure, chronic diseases (hypertension, dyslipidemia, diabetes mellitus, cancer, chronic lung disease, stroke, liver disease, arthritis, digestive disease, asthma).

CKD. Shi et al. conducted a cross-sectional study of the relationship between surrogate makers of insulin resistance and chronic kidney disease in Chinese hypertensive patients, which indicated that TyG index was associated with 48% increased risk of CKD and TyG index can be a predictor of CKD for hypertensive patients [18]. A cohort

study conducted by Xu et al. in Lanzhou, China showed that compared with the lowest quartile group, the CKD risk of the highest quartile group was 1.815 times than the lowest quartile group [15]. Conversely, Pan et al. found higher TyG index was significantly associated with increased risk of microalbuminuria (MAU) and ankle-

Table 3 Subgroup analysis of the association between TyG index and CKD.

Characteristics	TyG index				P value	P-interaction
	Q1 (<8.23)	Q2 (8.23–8.61)	Q3 (8.61–9.07)	Q4 (≥9.07)		
Sex						
Male	1.00 (Ref)	0.86 (0.62–1.20)	1.27 (0.94–1.72)	1.60 (1.11–1.66)	0.002	0.277
Female	1.00 (Ref)	1.13 (0.84–1.52)	1.15 (0.86–1.54)	1.44 (1.08–1.92)	0.012	
Age, years						
<60	1.00 (Ref)	1.04 (0.80–1.36)	1.09 (0.83–1.43)	1.27 (0.98–1.65)	0.068	0.173
≥60	1.00 (Ref)	1.06 (0.80–1.40)	1.11 (0.84–1.47)	1.31 (1.01–1.73)	0.047	
Place of residence						
Urban	1.00 (Ref)	1.52 (1.06–2.19)	1.56 (1.09–2.24)	1.93 (1.37–2.72)	<0.001	0.562
Rural	1.00 (Ref)	0.92 (0.73–1.16)	0.98 (0.77–1.23)	1.08 (0.86–1.37)	0.474	
BMI, kg/m ²						
<24	1.00 (Ref)	1.06 (0.85–1.32)	1.03 (0.82–1.31)	1.24 (0.98–1.57)	0.126	0.097
≥24	1.00 (Ref)	1.06 (0.71–1.58)	1.23 (0.85–1.78)	1.34 (0.94–1.91)	0.055	
Smoking						
No	1.00 (Ref)	1.03 (0.79–1.34)	1.13 (0.87–1.47)	1.30 (1.01–1.68)	<0.001	0.911
Yes	1.00 (Ref)	1.06 (0.80–1.41)	1.02 (0.76–1.36)	1.16 (0.86–1.56)	0.403	
Drinking						
No	1.00 (Ref)	1.04 (0.80–1.34)	1.02 (0.79–1.32)	1.22 (0.94–1.58)	0.150	0.706
Yes	1.00 (Ref)	1.05 (0.78–1.41)	1.19 (0.88–1.60)	1.28 (0.95–1.72)	0.077	
Hypertension						
No	1.00 (Ref)	1.00 (0.81–1.25)	1.14 (0.92–1.42)	1.26 (1.01–1.57)	0.022	0.322
Yes	1.00 (Ref)	1.20 (0.808–1.80)	0.89 (0.58–1.36)	1.21 (0.81–1.81)	0.639	
Education level, n (%)						
Below primary school	1.00 (Ref)	1.16 (0.82–1.64)	0.90 (0.62–1.29)	1.08 (0.74–1.56)	0.936	0.154
Primary school	1.00 (Ref)	1.06 (0.78–1.43)	1.32 (0.98–1.76)	1.26 (0.93–1.71)	0.062	
Middle school	1.00 (Ref)	0.74 (0.48–1.16)	0.79 (0.50–1.25)	1.15 (0.76–1.74)	0.459	
High school or above	1.00 (Ref)	1.55 (0.76–3.17)	1.58 (0.77–3.23)	2.14 (0.68–2.72)	0.034	

In the multivariate models, confounding factors such age, sex, place of residence, education level, smoking, drinking, systolic blood pressure, physical activity, chronic diseases (dyslipidemia, chronic lung disease and stroke) and medications (anti-hypertensive and anti-dyslipidemic) were included unless the variable was used as a subgroup variable.

brachial index (ABI) but not CKD (OR = 1.02, 95% CI: 0.87–1.21, $P = 0.783$) [25]. Through a cohort study based on the CHARLS, we found that compared with the lowest quartile group, the population with the highest quartile group increased the risk of CKD by 30%, there is a significant positive correlation between TyG index and CKD. Synthesizing the above research, the reasons for the inconsistent results are that due to the different types of study designs (the ability to infer causality in cross-sectional studies is low), the number of participants is uneven between studies and there are large differences in the inclusion criteria, age, gender, and race of participants.

Meta-analysis is a statistical method for comparing and synthesizing multiple research results with the same research purpose and independent research results, which can improve the statistical power of the original results, solve the inconsistency of the research results and explore the causes, so as to improve the effect estimates. We confirmed that individuals with a higher TyG index had an increased risk of CKD among participants by performing a meta-analysis of 12 previous studies and the present study. Our meta-analysis findings were consistent with previous studies that TyG index was significantly associated with CKD [14–24]. In summary, these findings suggested that TyG index might be a valuable predictor of CKD.

IR is prevalent in a variety of metabolic-related diseases. For IR, previous studies have proposed the TyG index, and

a few prospective studies have indicated that the TyG index was associated with new-onset diabetes [32], hypertension [33] and cardiovascular events [34]. Our cohort study and meta-analysis further expanded and confirmed that the TyG index is associated with CKD. The TyG index may be used as an independent predictor of CKD risk and it is clinical interest for clinicians to use TyG index as an predictive tool to find individuals with high risk of CKD.

At present, the exact mechanism between the TyG index and CKD is not clear. The TyG index is composed of lipid-related factors and glucose-related factors, which are closely related to IR. IR is a significant marker of metabolic syndrome symptoms such as diabetes, obesity, hypertension, and dyslipidemia (hypertriglyceridaemia and decreased high-density lipoprotein (HDL)) [35–37]. Hyperinsulinemia induces renal vasodilation leading to impairment of glomerular hyperfiltration and endothelial function thereby increasing proteinuria excretion [38–41]. Increased relaxation of nephrons leads to nephron loss, resulting in glomerular hypertension, which leads to glomerulosclerosis and renal impairment [15,42]. Meanwhile, the relationship between IR and CKD can be explained by biological mechanisms such as inflammation, metabolic acidosis and oxidative stress. IR can induce an imbalance of glucose metabolism to provoke inflammation, and at the same time lead to hyperglycemia which leads to metabolic acidosis to stimulate the excessive secretion of endothelin-

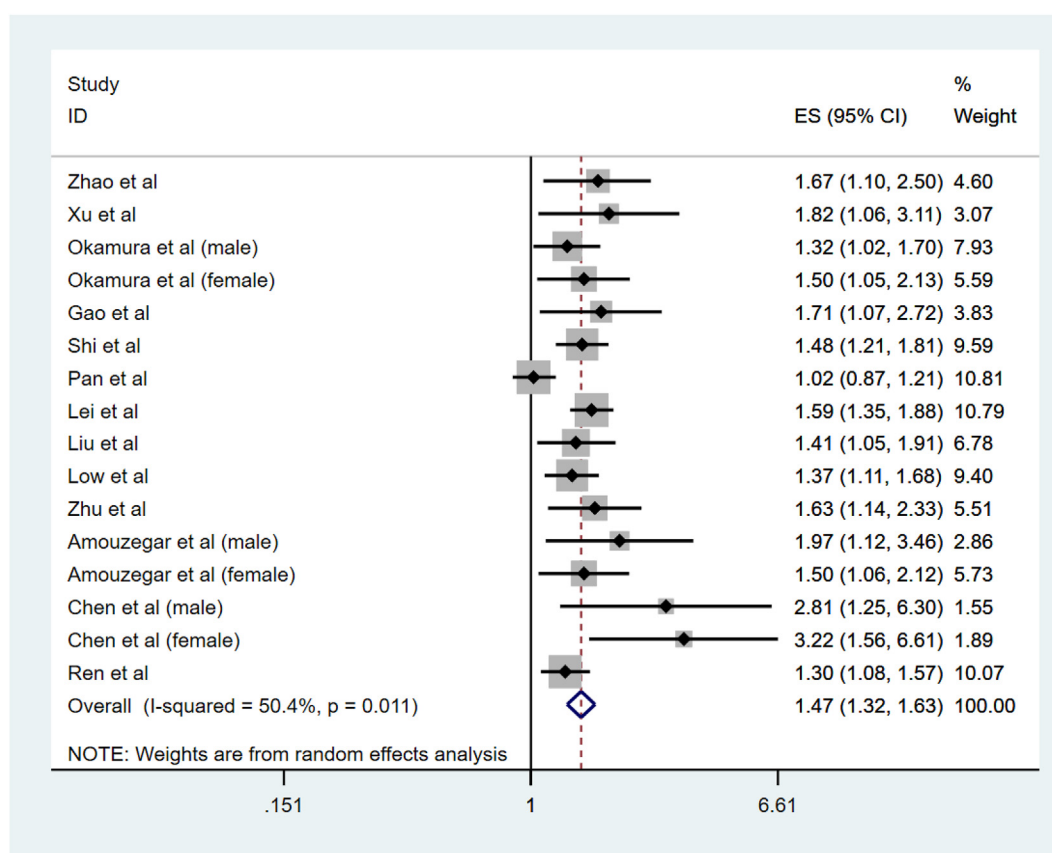


Figure 2 Meta-analysis of relative risks (RRs) and 95% CIs for the highest vs. lowest TyG index categories and CKD.

1 and aldosterone, resulting in tubulointerstitial damage. Furthermore, IR induces overproduction of reactive oxidative stress (ROS) by activating the mitochondrial electron transport chain, resulting in renal tissue fibrosis.

This study has the following main advantages. It is a large-scale longitudinal study, which can better explore the correlation between TyG index and the risk of CKD and the corresponding causal relationship. Furthermore, this study is the first meta-analysis of observational studies concerning the association between the TyG index and CKD based on the most comprehensive literature search to date. Nonetheless, several limitations of this study should be taken into consideration. First, the participants in this study are all middle-aged and elderly Chinese residents over the age of 45, and there is currently no reliable data on TyG index in the younger population. In addition, whether the associations identified in this study are applicable to other ethnic groups is still unknown, and future research summaries should attempt to further generalize to another population. Second, some potential confounding factors might influence the results were not considered in the current research. Third, higher heterogeneity in meta-analysis may affect the reliability of the results and potential publication bias might influence the findings.

In conclusion, we demonstrated that TyG index can be a predictor of incident CKD based on the CHARLS in China and higher TyG index was associated with increased risk of CKD, independently of established risk factors.

Authors' contributions

All authors contributed to the study conception and design. Design: Xiaowei Zheng, Xiao Ren; data analysis/interpretation: Xiao Ren, Minglan Jiang; statistical analysis: Xiao Ren; Longyang Han; supervision or mentorship: Xiaowei Zheng. Each author contributed important intellectual content during manuscript drafting or revision and agrees to be personally accountable for the individual's own contributions and to ensure that questions pertaining to the accuracy or integrity of any portion of the work, even one in which the author was not directly involved, are appropriately investigated and resolved, including with documentation in the literature if appropriate.

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Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.numecd.2023.03.026>.

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